

Thirty use cases, no accelerator

DG32-LITE single 50 MHz domain — the AI tasks, models and applications the part runs with no attention engine in the path.

USE CASES

30

MODEL TYPES

19

ALL RUN AT

≥ 79 Hz

MOST RUN AT

> 1 kHz

ACCELERATOR

None

Contents

Every entry links to its slide.

01	The envelope	→	02	Algorithms on a scalar core	→
03	Model catalogue	→	04	Feature extraction	→
05	Use cases · rotating machinery	→	06	Use cases · electrical and power	→
07	Use cases · control, motion and sensing	→	08	Use cases · slower-rate and sequence	→
09	What the sensing supports	→	10	Basis	→

What you are working with

Every number on the following slides derives from these four.

SCALAR THROUGHPUT 12.5 MMAC/s RV32IM at 50 MHz, int8, well blocked	MODEL BUDGET 16.5 KB 29.5 KB with the runtime moved to mask ROM	FREE CYCLES 82 % Assuming a 10 kHz field-oriented control loop	HARDWARE HELP CORDIC sin, cos, atan2, magnitude — already in the loop
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What the missing engine costs

Speed on one operator, not capability. The attention kernel at 64/32/32 is 0.26 MMAC — about 21 ms in software. Nothing on the following slides needs it: the algorithms that suit a scalar core are 30–300× cheaper than a transformer at equal accuracy on these tasks.

The budget at each rate

MAC budget per inference is $12.5 \times 10^6 \div \text{rate}$. At 10 Hz that is 1.25 MMAC; at 100 Hz, 0.125 MMAC; at 1 kHz, 12.5 kMAC. The second gate is memory: weights plus peak activation must clear 16.5 KB.

Every use case here clears both

Thirty use cases, worst case 10.3 ms and 20 KB. Twenty-four of the thirty run above 1 kHz. The constraint that actually binds is not compute or memory — it is the analog front end, covered on slide 09.

Where a scalar core is strong

The absence of a MAC array reverses the usual ranking. Comparisons and table lookups cost almost nothing here.

Tree ensembles are the best accuracy per cycle on this part

A 100-tree random forest at depth 8 is roughly 800 comparisons and **no multiplies at all** — 0.06 ms, supporting rates above 1 kHz. A neural network of comparable accuracy on bearing-fault or power-quality classification costs 50–500× more cycles.

This is not a compromise made for the hardware. On the CWRU bearing benchmark a random forest over five time-domain features reaches 95.6 %, and cutting from nine features to five costs 0.1 %. A review of 42 papers finds SVM at 95–100 % against deep methods at 97–100 % — statistically indistinguishable.

Model capacity is not the bottleneck on these tasks. Signal conditioning is.

The ordering that follows

First choice — tree ensembles, linear discriminants, PCA with Hotelling T^2 , Mahalanobis distance. Sub-millisecond, kHz-capable, inspectable in the field.

Second choice — small dense networks and autoencoders, where a nonlinear boundary genuinely earns its place. Still sub-millisecond.

Reach for last — recurrent and convolutional models. A GRU at 2.95 ms and a 1D-CNN at 10.3 ms both fit, but only where the raw waveform carries something hand-built features miss.

Design rule. Spend effort on band selection, envelope demodulation and operating-condition normalisation before spending it on model capacity.

The nineteen that fit

Inference cost only — feature extraction is costed separately on the next slide. Memory is weights plus working state.

MODEL	ROLE	CYCLES	INFERENCE	MEMORY	MAX RATE
Random forest, 100×d8	Primary classifier	3,200	0.06 ms	20 KB	>1 kHz
Gradient boosting, 200×d4	Primary classifier	3,200	0.06 ms	12 KB	>1 kHz
Isolation forest	Unlabelled anomaly detection	3,200	0.06 ms	8 KB	>1 kHz
LDA	Primary classifier	128	0.00 ms	0.7 KB	>1 kHz
Logistic regression	Primary classifier	128	0.00 ms	0.7 KB	>1 kHz
Naive Bayes	Regime identification	512	0.01 ms	0.5 KB	>1 kHz
PCA + Hotelling T²	Anomaly detection — the condition-monitoring standard	1,024	0.02 ms	2 KB	>1 kHz
Mahalanobis score	Anomaly score, single scalar	4,096	0.08 ms	2 KB	>1 kHz
One-class SVM	Anomaly detection with a decision boundary	12,800	0.26 ms	6.9 KB	>1 kHz
k-NN, 200 prototypes	Field-adaptable classifier	25,600	0.51 ms	6.4 KB	>1 kHz
Nearest centroid	On-device baselining, no backward pass	1,024	0.02 ms	1 KB	>1 kHz
Gaussian mixture	Regime identification	2,048	0.04 ms	2 KB	>1 kHz
HMM, 8 state	Sequence and state tracking	3,072	0.06 ms	2 KB	>1 kHz
Extended Kalman filter	Observer — sensorless FOC, thermal	2,400	0.05 ms	2 KB	>1 kHz
MLP 32-16-8-4	Nonlinear classifier / regressor	2,688	0.05 ms	0.7 KB	>1 kHz
MLP autoencoder	Unsupervised drift detection	22,016	0.44 ms	5.5 KB	>1 kHz
MLP 128-64-32-8	Regression at the dense-network ceiling	41,984	0.84 ms	10.5 KB	970 Hz
GRU, 16 units	Short sequence modelling	147,456	2.95 ms	1.6 KB	270 Hz
1D-CNN 8/16/32	Raw-waveform classification	512,000	10.24 ms	14 KB	80 Hz

Feature extraction, costed

Every use case pairs one or two of these with one model. The CORDIC unit does Park, Clarke, magnitude and atan2 in hardware, so the diagnostic front end reuses what the control loop already computes.

STAGE	CYCLES	TIME	WHAT IT YIELDS
256-point real FFT	30,720	0.61 ms	Band energies, spectral peaks, harmonic ratios
512-point real FFT	69,120	1.38 ms	Finer resolution for gear-mesh and sideband work
Envelope demodulation	5,000	0.10 ms	CORDIC magnitude, 20 iterations — the bearing-fault front end
Goertzel, 8 bins	6,144	0.12 ms	Narrowband detection at predicted fault frequencies
Statistical moments	7,000	0.14 ms	RMS, kurtosis, crest factor, skew — ISO 20816 baseline
Broadband RMS velocity	3,000	0.06 ms	ISO 20816 severity zone A/B/C/D
Haar wavelet, 256 samples	15,000	0.30 ms	Time-frequency features for transient faults
Park / Clarke transform	120	0.00 ms	CORDIC, already computed by the control loop
Decimation and band-pass	4,000	0.08 ms	IIR biquads ahead of envelope or raw-waveform models

Why this matters more than the model

A published feature-ranking study across twelve features found demodulated features evaluated **at the known fault frequency** outrank raw statistical moments by a factor of 4-5. Hilbert-Huang amplitude at the outer-race frequency scored 225.9; peak-to-peak scored 51.8.

That is the whole argument for spending cycles on envelope demodulation and band selection rather than on a larger classifier. The envelope chain is also cheap: band-pass, rectify, low-pass, decimate to 1-2 kS/s, then a modest FFT. Do the decimation before buffering and the working set never exceeds a few kB.

One caution. Kurtosis is non-monotonic — it peaks for incipient localised defects then falls as damage spreads and the signal becomes Gaussian. RMS is the opposite: stable and monotonic, but blind to early faults. Trend both; never alarm on kurtosis alone.

Rotating machinery

USE CASE	WHAT IT DETECTS	SENSING	FEATURES	MODEL	MEMORY	LATENCY	MAX RATE
Bearing fault classification	Inner race, outer race, ball and cage defects	Accelerometer, ≥5 kHz bandwidth	envelope + FFT-256 + moments	Random forest, 100×d8	20 KB	0.92 ms	890 Hz
Bearing severity trending	ISO 20816 zone A/B/C/D from broadband RMS velocity	Accelerometer	decimate + RMS	LDA	0.7 KB	0.14 ms	>1 kHz
Gearbox and gear-mesh faults	Tooth damage, mesh-frequency sideband growth	Accelerometer, order-tracked on encoder	envelope + FFT-512 + moments	Random forest, 100×d8	20 KB	1.69 ms	480 Hz
Pump cavitation and dry-run	Separates cavitation, dry running and normal load	Phase current + pressure over SPI	FFT-256 + moments	Gradient boosting, 200×d4	12 KB	0.82 ms	>1 kHz
Fan and blower imbalance	Blade damage, 1× and 2× amplitude ratio	Accelerometer + tachometer	FFT-256 + moments	LDA	0.7 KB	0.76 ms	>1 kHz
Compressor valve faults	Valve leakage and flutter against crank angle	Accelerometer + crank reference	envelope + FFT-256	Random forest, 100×d8	20 KB	0.78 ms	>1 kHz
Belt slip and misalignment	Dual-encoder differential, or vibration harmonic ratio	Dual encoder or accelerometer	FFT-256	Logistic regression	0.7 KB	0.62 ms	>1 kHz
Shaft misalignment and looseness	1×, 2×, 3× harmonic pattern	Accelerometer	FFT-256 + moments	Random forest, 100×d8	20 KB	0.82 ms	>1 kHz

Electrical and power

USE CASE	WHAT IT DETECTS	SENSING	FEATURES	MODEL	MEMORY	LATENCY	MAX RATE
Broken rotor bar detection	Twice-slip-frequency sidebands, $f_1(1\pm2k_s)$	Phase current, ≥ 12 -bit with notch	Goertzel	LDA	0.7 KB	0.13 ms	>1 kHz
Air-gap eccentricity	$f_1 \pm m \cdot f_r$ sidebands from mixed eccentricity	Phase current, ≥ 12 -bit	Goertzel	LDA	0.7 KB	0.13 ms	>1 kHz
Stator inter-turn short	Negative-sequence current index against baseline	Three-phase current + voltage	Park + RMS	Mahalanobis score	2 KB	0.14 ms	>1 kHz
Phase loss and current unbalance	Sequence-component imbalance	Phase current, already sampled	Park	Logistic regression	0.7 KB	<0.01 ms	>1 kHz
Arc-fault and partial discharge	Early insulation breakdown signature	HF current transformer + envelope	envelope + FFT-256	One-class SVM	6.9 KB	0.97 ms	840 Hz
Power-quality events	Sag, swell, harmonic distortion, flicker	Line voltage and current, 64 cycles	Goertzel	LDA	0.7 KB	0.13 ms	>1 kHz
Battery state-of-health	SoH and SoC from charge-cycle shape	Slow V / I / T sequence	moments	MLP 32-16-8-4	0.7 KB	0.19 ms	>1 kHz
Winding thermal estimation	Virtual temperature sensor from current history	Current history + ambient	RMS	Extended Kalman filter	2 KB	0.11 ms	>1 kHz

Control, motion and sensing

USE CASE	WHAT IT DETECTS	SENSING	FEATURES	MODEL	MEMORY	LATENCY	MAX RATE
Sensorless rotor position	EKF flux observer, removes the position sensor	Phase current, CORDIC-assisted	Park	Extended Kalman filter	2 KB	0.05 ms	>1 kHz
Learned sensor plausibility	Encoder vs Hall vs back-EMF consistency	Existing peripherals, no added BOM	Park	Nearest centroid	1 KB	0.02 ms	>1 kHz
Operating-mode classification	Which duty regime the machine is in	Encoder, Hall, current, temperature	RMS	Gaussian mixture	2 KB	0.10 ms	>1 kHz
Duty-cycle and state tracking	Start counts, dwell times, regime transitions	Existing peripherals	–	HMM, 8 state	2 KB	0.06 ms	>1 kHz
Adaptive friction compensation	Learned cogging and friction feedforward	Encoder + current	Park	MLP 32-16-8-4	0.7 KB	0.06 ms	>1 kHz
Kickback and stall detection	Sudden load transient classification	Phase current	RMS	Gradient boosting, 200×d4	12 KB	0.12 ms	>1 kHz
Load estimation and torque ripple	Torque ripple harmonics from current	Phase current	FFT-256	Logistic regression	0.7 KB	0.62 ms	>1 kHz
Multivariate anomaly scoring	Single scalar novelty score, raised as an IRQ	Any fused sensor set	moments	Mahalanobis score	2 KB	0.22 ms	>1 kHz

Slower-rate and sequence

USE CASE	WHAT IT DETECTS	SENSING	FEATURES	MODEL	MEMORY	LATENCY	MAX RATE
Remaining-useful-life regression	Trend extrapolation over long feature history	64 hourly feature snapshots	–	MLP 128-64-32-8	10.5 KB	0.84 ms	970 Hz
Unsupervised drift detection	Reconstruction error against learned healthy manifold	Any fused sensor set	moments	MLP autoencoder	5.5 KB	0.58 ms	>1 kHz
Short-horizon forecasting	Predicts a slow process variable a few steps ahead	Any slow scalar, 64-step history	–	GRU, 16 units	1.6 KB	2.95 ms	270 Hz
Raw-waveform fault classification	End-to-end from the sampled window, no hand features	Accelerometer or current	decimate	1D-CNN 8/16/32	14 KB	10.32 ms	79 Hz
Novelty detection without labels	Isolation-forest outlier score	Any fused sensor set	moments	Isolation forest	8 KB	0.20 ms	>1 kHz
Per-machine baselining	On-device prototype update, no backward pass	Any fused sensor set	moments	k-NN, 200 prototypes	6.4 KB	0.65 ms	>1 kHz

What the sensing supports

Compute is not the binding constraint on these thirty use cases. The analog front end is.

The current-based use cases need more than 8 bits

ISO 13373-2 gives usable dynamic range as **$D = 6(N-1)$ dB**, so eight bits yields 42 dB. ISO 20958 states rotor-bar sidebands sit 100 to 1,000 times below the fundamental — **−40 to −60 dBc**. The entire diagnostic range sits at or below the quantisation floor. Broken rotor bar, eccentricity and inter-turn short detection need either analog fundamental suppression and gain ahead of the converter, or 12–16 bits. A second constraint compounds it. Sideband separation is $2 \cdot s \cdot f_1$ — roughly 3 Hz at full load, 0.5 Hz at light load — so these use cases need **30–100 second records at stable load and speed** with 0.01–0.05 Hz resolution. A 2^{20} -point FFT is 8 MB and does not fit. Use Goertzel evaluated at the predicted sideband frequencies, which is why the electrical use cases on slide 06 are costed that way and not with an FFT.

Tier 1 — one accelerometer

The highest-value addition. Bearings are 44 % of motor failures and are the weakest fault class for current-only sensing; rotor bars, where current is strongest, are 8–10 %. Needs **≥ 5 kHz usable bandwidth** — many low-cost MEMS parts roll off at 1–3 kHz, below the bearing resonance band, which makes envelope analysis useless. Mounting matters as much as the part: stud mounting per ISO 13373-1, not magnet or adhesive.

Tier 2 — three-phase current and voltage

Unlocks negative-sequence detection of stator inter-turn shorts, instantaneous power signature analysis and Park's vector per ISO 20958 Annex A. Needs a multi-channel converter; the part currently has one differential channel.

What works today, unchanged

Everything on slide 08 and most of slide 09 — vibration-based classification, sensorless observation, plausibility checking, regime identification, anomaly scoring and duty tracking — runs on the sensing already present, because they depend on relative change rather than absolute small-signal resolution.

Basis

Read this before quoting any figure elsewhere.

From the design

Core configuration, memory sizes, peripheral set and the CORDIC capability come from the DG32-LITE block diagrams dated 10 September 2026 and the consolidated capability assessment of 11 September 2026. The 16.5 KB and 29.5 KB model budgets and the 12.5 MMAC/s scalar rate are taken from that assessment.

Derived

Every latency, memory figure and maximum rate in this deck is calculated — feature-extraction cycles plus model cycles at 4 cycles per int8 multiply-accumulate, against 82 % of a 50 MHz core. **That constant is back-solved from the design documents and has not been measured on silicon.** One cycle-accurate measurement on run R firmware relocates every rate here.

Third-party, verified 11 Sept 2026

CWRU random-forest accuracy and the 42-paper review, the twelve-feature ranking study, the kurtosis non-monotonicity result, and the ISO signal-processing limits were checked against primary sources — IEEE and MDPI papers, the PHM Society proceedings, and the ISO catalogue entries for 13373-1, 13373-2, 20958 and 20816-3.

On accuracy expectations

Do not carry published bearing-benchmark accuracy into a specification. Forty of forty-one reviewed CWRU studies used splits vulnerable to data leakage; on a leakage-free bearing-wise split one classifier fell from 85.8 % to 69.5 %. Realistic accuracy on unseen physical bearings is **65–80 %**. Smith and Randall further showed a meaningful fraction of CWRU records are not diagnosable by correct physics at all, which is why 99 % figures on that dataset should be treated as artefacts.

What is not claimed

No workload here has been compiled for or measured on DG32-LITE, which has not completed place-and-route. No inference result carries a safety integrity claim — that architecture places the classifier in an advisory role with a lockstepped deterministic monitor holding the trip limits. Model choices are the appropriate algorithm for each task on this core, not trained and validated implementations.